

Minneapolis Terminal Radar Approach Control – Satellite Towers LOA**(ANE, FCM, MIC, STP)****LETTER OF AGREEMENT****EFFECTIVE: 11/30/2025****SUBJECT: TERMINAL OPERATIONS**

1. **PURPOSE.** To prescribe procedures for the control of IFR and VFR air traffic at Anoka, Flying Cloud, Crystal, St. Paul Airports.
2. **CANCELLATION.** Initial Release.
3. **SCOPE.** This agreement between Minneapolis Terminal Radar Approach Control (M98) and its outlying Satellite (ANE, FCM, MIC, STP) ATCTs establishes responsibilities and procedures for handling IFR and VFR air traffic at the ANE, FCM, MIC, and STP Airports. These procedures are supplementary to FAA Order JO 7110.65, Air Traffic Control.
4. **DEFINITIONS.** Hereafter certain terms in this LOA shall be defined by abbreviations.

ATCT or ATCTs shall be defined as ANE, FCM, MIC, and STP ATCTs.

5. RESPONSIBILITIES.

- a. M98 must be responsible for providing approach control services to IFR arrivals, departures, and practice instrument approaches to ATCT airports in accordance with procedures contained herein.
- b. ATCTs must be responsible for designated IFR and SVFR operations in accordance with the procedures contained herein.
- c. Deviations from the procedures contained herein must be approved only after coordination between the facilities is effected which specifically defines responsibility and intent.

6. PROCEDURES.

- a. M98 must:
 - 1) Departures.
 - (a) Issue departure releases via the Run-Down Tab List. Verbally coordinate nonstandard headings and altitudes. (Standard headings and keyboard entries are listed in Attachment 2.)

2) Arrivals.

(a) Ensure that full data blocks for arriving traffic contain the following data, unless verbal coordination with the appropriate ATCT is effected:

1. Scratch pad entry data as listed in Attachment 1.
2. Type aircraft in field four.

3) Missed Approaches.

(a) Issue IFR aircraft planning a missed approach the following instructions:

ANE: Heading 360° and 2,500 ft.

FCM: Heading 230° and 2,500 ft.

MIC: Heading 320° and 2,500 ft.

STP: Heading 060° and 2,500 ft.

(b) Advise ATCTs of heading and altitude assigned to IFR and VFR aircraft executing a missed approach if different from (a) above.

4) Overflights must be authorized to transit ATCT airspace at or above 3000 ft. MSL without prior coordination. Overflight traffic not displayed (automatically or manually) to ATCT CTRDs must be verbally coordinated:

b) ATCTs must:

1) Departures.

(a) Assign the following initial altitudes to IFR departures from ATCT airports:

Airport	Runway(s)	Altitude
ANE	All	3,000
FCM	10L/R, 36	2,500
	28L/R, 18	3,000
MIC	All	3,000
STP	13, 14	2,500
	32	3,000

(b) Use Run-Down Tab List procedures in Attachment 2 to obtain IFR departure releases from M98.

- (c) Place aircraft in Run-Down Tab List, in the order they will depart, using STARS entries listed in Attachment 2.
- (d) Verbally coordinate any changes that have been made in the Run-Down Tab List whether or not the flash has been recalled.
- (e) Verbally coordinate all departures from non-advertised runways and any non-standard heading and/or altitude.

2) Arrivals

- (a) FCM may change an aircraft's runway (side-step or circle) to any non-opposite direction runway without coordination. FCM assumes separation responsibility with any preceding and succeeding aircraft.
- (b) STP is authorized to interchange Runway 31 with 32 or Runway 13 with 14 when issuing landing instructions.

3) Missed Approaches.

- (a) ATCTs must instruct any IFR aircraft executing an unplanned missed approach to fly the following headings and altitudes:
 - ANE: Heading 360° and maintain 2,500 ft.
 - FCM: Heading 230° and maintain 2,500 ft.
 - MIC: Heading 320° and maintain 3,000 ft.
 - STP: Heading 060° and maintain 2,500 ft.
- (b) Advise M98 when an IFR aircraft executes an unplanned missed approach.

7. ATTACHMENTS.

Attachment 1 Scratch Pad Entries

Attachment 2 Run-Down Tab List Procedures

Dhruv Kalra
Air Traffic Manager
VATSIM Minneapolis ARTCC

Lance Harry
Deputy Air Traffic Manager
VATSIM Minneapolis ARTCC

Attachment 1: Scratch Pad Entries

- 1) Scratch Pad 1 must have airport of intended landing for IFR or VFR aircraft.

Example-

STP, FCM, ANE, MIC, LVN, 21D, SGS

- 2) Scratch Pad 2

- a) 1st character indicates type of instrument approach for IFR and VFR aircraft.
- b) 2nd/3rd character(s) indicates how the approach will terminate.
- c) VV – Aircraft has been cleared for a visual approach, and the pilot has assumed responsibility for separation (including wake turbulence separation) from the preceding aircraft. (Note: the scratchpad will look like VV+, the + is just an indication that the scratchpad is time-sharing).
- d) The following approach abbreviations must be entered in scratch pad 2:

1 st Character – TYPE of APPROACH	2 nd and 3 rd Character – Full Stop or Missed Approach
I = ILS	F = Full Stop
G = RNAV (GPS)	M = Missed Approach
L = Localizer	PM = Published Missed
N = NDB	VV = Pilot provided visual separation
V = VOR	

Example – N12345 requesting ILS 30 to a Published Missed at LVN: STARS entry will be <+IPM, slew, enter>.

Other Approved Entries

Air Work	A/W
Holding	HLD
Photo Flight	PIX
Holding at GEP	GEP
Holding at FGT	FGT
Tour	2ER

Attachment 2: Run-Down Tab List Procedures

Standard IFR departure headings and altitudes:

Airport	Runway(s)	Heading(s)	Altitude(s)
ANE	18	Right turn 270° clockwise to 090°	3,000
	9, 27, 36	270° clockwise to 090°	
FCM	28L, 28R, 18	180° clockwise to 300°	3,000
	10L, 10R, 36		2,500
MIC	All	320°	3,000
STP	32	030° clockwise to 045°	3,000
	14, 13	010° clockwise to 080°	2,500

STARS Entries:

Assume that commas in STARS entries represent a space or new line in STARS. Shift+F3 may be used in lieu of F13 for keyboards that do not have an F13 key.

- a. ATCT requests a release via the Run-Down Tab List:

Airport	Format	STARS Entry
ANE	Departure direction (N, S, W, or E), Heading <i>Note: N: Runway 36</i> <i>S: Runway 18</i> <i>W: Runway 27</i> <i>E: Runway 09</i>	F13, ACID, (N, S, W, or E) (270-090) <enter>
FCM	Departure direction, heading <i>Note: N: Runway 36</i> <i>S: Runway 18</i> <i>W: Runway 28</i> <i>E: Runway 10</i>	F13, ACID, (N, S, W, or E) (180-300) <enter>
MIC	Standard heading	F13, ACID, (320) <enter>
STP	Standard heading	F13, ACID, (010-080) <enter>

- b. M98 Issue release: *F13*, *<list ID>*, *ACID*, *<enter>*
- c. ATCT change departure sequence: *F13*, *ACID*, *<new sequence #>*, *<enter>*
- d. ATCT cancel release request: *F13*, *ACID*, *<enter>*
- e. ATCT or M98 move the Run-Down Tab List: *F7*, *<list ID>*, *<slew>*
- f. ATCT or M98 quicklook Run-Down Tab List header: *F7*, *<list ID>*, *E*, *<enter>*

M98 Run-Down Tab List ID Quick Reference:

ATCT	List ID
ANE	PO
FCM	PF
MIC	PM
STP	PP